Econ 812 Recitation 10 Notes

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1 A Growth Model with Externalities June 22

There is a continuum of households with representative household preferences over consumption given in equation 1.

$$\sum_{t=0}^{\infty} \beta^t 1/2 C_t^{1/2} \tag{1}$$

Where C_t is consumption. The household has 1 unit of time that it supplies inelastically in the labor market. Capital evolves according to

$$K_{t+1} = I_t$$

with $K_0 > 0$ given. The household's budget constraint is

$$C_t + I_t = r_t K_t + w_t + \pi_t$$

where π_t is any profit earned by the firms. There is a unit continuum of firms. The representative firm has a production function given by

$$y_t = \Gamma A_t k_t^{\alpha} n_t^{1-\alpha}$$

with $0 < \alpha < 1$. The firm takes A_t as given. The TFP term At depends on the aggregate production $A_t = Y_t^{\theta}$ with $\theta \geq 0$,$\theta \neq 1$ In a symmetric equilibrium $y_t = Y_t$.

1.1 Part A

Define a symmetric competitive equilibrium for this economy. [5 pts]

A symmetric competitive equilibrium is a set of prices $P = \{w_t, r_t\}$, a plan for the firm $F = \{k_t, n_t, \pi_t\}$ a plan for the households $H = \{K_t, I_t, C_t, N_t\}$ subject to given prices and an initial amount of capital, H solves the household's problem, F solves the firm's problem, goods markets clear $Y_t = C_t + I_t$, capital markets clear $k_t = K_t$ and labor markets clear $n_t = N_t$

1.2 Part B

Suppose $\theta = 1 - \alpha$. Under what conditions on model parameters do we a balanced growth path with positive growth. Under these conditions, solve for the competitive equilibrium. At what rates do consumption, wages, and output grow? [15 pts]

Step 1: Set up the HH problem

$$\max_{c_t, k_{t+1}} \sum_{t=0}^{\infty} \beta^t [1/2 c_t^{1/2} + \lambda_t [r_t k_t + w_t - k_{t+1} - c_t]]$$

Step 2: Taking FOCs

$$c_t : 1/4 c_t^{-1/2} = \lambda_t$$

$$k_{t+1} : \lambda_t = \beta \lambda_{t+1} r_{t+1}$$

$$\lambda_t : r_t k_t + w_t = k_{t+1} + c_t$$

Step 3: Solve for the EE

$$1/4 c_t^{-1/2} = \beta 1/4 c_{t+1}^{-1/2} r_{t+1} \implies c_t^{-1/2} = \beta c_{t+1}^{-1/2} \alpha \Gamma A_{t+1} k_{t+1}^{\alpha - 1} \tag{2}$$

Here we know the ratio $A/k^{1-\alpha}$ is constant.

Step 4: Use other equations and definitions from problem to solve for growth rate. (Here we know $A_t = y_t^{1-\alpha}$)

$$y_t = \Gamma y_t^{1-\alpha} k_t^{\alpha}$$

$$y_t^{\alpha} = \Gamma k_t^{\alpha}$$

$$\Gamma A_t k_t^{\alpha} = \Gamma^{1/\alpha} k_t$$

$$\frac{A_t}{k_t^{1-\alpha}} = \Gamma^{1/\alpha - 1}$$

Plug back into EE:

$$\gamma_c^{1/2} = \beta \alpha \Gamma (\Gamma^{1/\alpha - 1})$$
$$\gamma_c = \beta^2 \alpha^2 \Gamma^{2/\alpha} > 1$$

Step 5: Use firm's production function to argue why $\gamma_y = \gamma_k$

$$\gamma_y = \frac{Y_{t+1}}{Y_t} = \frac{\Gamma A_{t+1} k_{t+1}^{\alpha}}{\Gamma A_t k_t^{\alpha}} = \gamma_A \gamma_k^{\alpha} \tag{3}$$

From the definition of A_t

$$\gamma_A := \frac{A_{t+1}}{A_t} = (\frac{Y_{t+1}}{Y_t})^{1-\alpha} = \gamma_y^{1-\alpha} \tag{4}$$

Substitute 4 into 3 to find that $\gamma_y = \gamma_k$.

Step 6: Say that by the BC and TVC $\gamma_c = \gamma_y = \gamma_k$.

1.3 Part C

Suppose $\theta = 1 - \alpha$. Under what conditions on model parameters do we a balanced growth path with positive growth for the solution to the social planner. Under these conditions, solve for the balanced-growth path solution to the social planner and identify growth rates. [15 pts]

In the SPP, the social planner is able to internalize the externality, and so it can change the level of technology since it controls the level of output. Here, $Y_t = \Gamma A_t k_t^\alpha \implies Y_t = \Gamma Y_t^{1-\alpha} k_t^\alpha \implies Y_t = \Gamma^{1/\alpha} k_t$. This reduces to the Ak model.

Step 1: Set up SPP lagrangian

$$\max_{c_t, k_{t+1}} \sum_{t=0}^{\infty} \beta^t [\frac{1}{2} c_t^{1/2} + \lambda_t [\Gamma^{1/\alpha} k_t - c_t - k_{t+1}]]$$

Step 2: Taking FOCs

$$c_t : \frac{1}{4} c_t^{-1/2} = \lambda_t$$

$$\beta \Gamma^{1/\alpha} \lambda_{t+1} = \lambda_t$$

Plus RC and TVC.

Step 3: Solve for the EE

$$\frac{1}{4} c_t^{-1/2} = \beta \Gamma^{1/\alpha} \frac{1}{4} c_{t+1}^{-1/2}$$

$$(\frac{c_{t+1}}{c_t})^{1/2} = \beta \Gamma^{1/\alpha}$$

$$\gamma_c = \beta^2 \Gamma^{2/\alpha}$$

Step 4: Use other equations and definitions from problem to solve for growth rate. (Done)

Step 5: Use firm's production function to argue why $\gamma_y = \gamma_k$

$$\gamma_y = \frac{Y_{t+1}}{Y_t} = \frac{\Gamma A_{t+1} k_{t+1}^\alpha}{\Gamma A_t k_t^\alpha} = \gamma_A \gamma_k^\alpha \tag{5}$$

From the definition of A_t

$$\gamma_A := \frac{A_{t+1}}{A_t} = (\frac{Y_{t+1}}{Y_t})^{1-\alpha} = \gamma_y^{1-\alpha} \tag{6}$$

Substitute 4 into 3 to find that $\gamma_y = \gamma_k$.

Step 6: Say that by the BC and TVC $\gamma_c = \gamma_y = \gamma_k$.

1.4 Part D

How does the growth and level of income in the competitive equilibrium compare with that of the solution to the social planner's problem? Explain. [8 pts]

From SPP we have $\gamma_c = \beta^2\Gamma^{2/\alpha}$ and from CE we have $\gamma_c = \beta^2\alpha^2\Gamma^{2/\alpha}$. Since $0 < \alpha < 1$ we have that the CE is smaller than the SPP. This is what we saw from class, and stems from the fact that the SP is able to internalize the externality... that the $MPK > r$.

1.5 Part E

Is it possible to have economic growth under the social planner's problem and not in the competitive equilibrium? Explain. [8 pts]

For this situation to occur, we would need parameter values such that $\beta^2\Gamma^{2/\alpha} > 1$ and $\beta^2\alpha^2\Gamma^{2/\alpha} < 1$

2 Productive Government Spending June 21

Government spending $\{G_t\}$ is deterministic and exogenous. The government sets lump-sum taxes T_t to balance its flow budget constraint each period:

$$T_t = G_t$$

Suppose the representative household has preferences given by

$$\sum_{t=0}^{\infty} \beta^t u(C_t, N_t)$$

where

$$u(C_t, N_t; G_t) = \log C_t - N_t$$

where $\beta = 1/2$. The HH selects labor supply, consumption and investment to maximize utility s.t.

$$C_t + I_t = \Pi_t + R_t K_t + w_t N_t - T_t$$

and LOM for capital

$$K_{t+1} = I_t$$

The firm's production function is

$$Y_t = 4G_t K_t^{1/2} N_t^{1/2}$$

2.1 Part A

(5 points) Write the household's Lagrangean, first-order conditions and transversality condition.

$$\max_{C_t, N_t, K_{t+1}} \mathscr{L} = \sum_{t=0}^{\infty} \beta^t [\ln C_t - N_t + \lambda_t [R_t K_t + w_t N_t - T_t - C_t - K_{t+1}]]$$

FOCs:

$$C_t : 1/C_t = \lambda_t$$

$$N_t : 1 = \lambda_t w_t \implies w_t = C_t \tag{7}$$

$$K_{t+1} : \lambda_t = \beta \lambda_{t+1} R_{t+1} \implies C_{t+1}/C_t = \beta R_{t+1} \tag{8}$$

Plus the BC and TVC

2.2 Part B

(5 points) Write down the firm's first-order conditions. Intuitively explain how government spending affects labor demand and capital demand.

$$\max_{k_t, n_t} 4 G_t k_t^{1/2} n_t^{1/2} - w_t n_t - r_t k_t$$

FOCs:

$$k_t : 2 G_t (k_t/n_t)^{-1/2} = r_t \tag{9}$$

$$n_t : 2 G_t (k_t/n_t)^{1/2} = w_t \tag{10}$$

Here, government spending increases the price of factor inputs (r_t and w_t) which implies capital and labor is in higher demand.

2.3 Part C

(10 points) Suppose that steady state government spending G = 1. Solve for the steady state C, I, K, N, Y, w, R, T.

We have 8 unknowns and the following equations: 7,8,9,10, plus $K = I$,$Y = C + I + G$,$G = T$, and $C + I = RK + wN - T$. So we can solve!

From 8 we have $R = \beta^{-1}$, plugging into 9 we get $K/N = (\frac{2}{\beta^{-1}})^2$ From 10 and 7 we have $w = C = 2(\frac{2}{\beta^{-1}})$, then we also have $G = 1 = T$. The RC is a bit trickier

$$C + K = RK + wN - T$$

$$4/\beta^{-1} + K = \beta^{-1} K + 4/\beta^{-1} N - 1$$

Divide everything by N

$$4/\beta^{-1}/N + K/N = \beta^{-1} K/N + 4/\beta^{-1} - 1/N$$

$$1/N(4\beta+1) = 4\beta + (\frac{2}{\beta^{-1}})^2(\beta^{-1}-1)$$

$$N = \frac{4\beta+1}{4\beta + (\frac{2}{\beta^{-1}})^2(\beta^{-1}-1)}$$

Then can solve for $I = K$ because know K/N

$$K = I = (\frac{4\beta+1}{4\beta + (\frac{2}{\beta^{-1}})^2(\beta^{-1}-1)})(\frac{2}{\beta^{-1}})^2$$

2.4 Part D

Log-linearize the household's and firm's first-order conditions for N_t and K_{t+1}.

H.H.

$$\hat{w}_t = \hat{c}_t$$

$$1/c * c * \hat{c}_{t+1} + (-1)c/c^2 * c * \hat{c}_t = \beta r * \hat{r}_{t+1} \implies \hat{c}_{t+1} - \hat{c}_t = \hat{r}_{t+1}$$

Firm

$$2(k/n)^{1/2} G\hat{G}_t + Gk^{-1/2}n^{-1/2}k\hat{k}_t - Gk^{1/2}n^{-3/2}n\hat{n}_t = w\hat{w}_t$$

$$\hat{G}_t + 1/2\hat{k}_t - 1/2\hat{n}_t = \hat{w}_t$$

$$\hat{G}_t - 1/2\hat{k}_t + 1/2\hat{n}_t = \hat{r}_t$$

2.5 Part E

(15 points) Suppose we have log-linearized our model to attain

$$Ax_{t+1} = Bx_t + Cv_t$$

where $x_t = \begin{bmatrix} \hat{K}_t \\ \hat{C}_t \end{bmatrix}$ and $v_t = \begin{bmatrix} \hat{G}_{t+1} \\ \hat{G}_t \end{bmatrix}$. Let $J = \begin{bmatrix} 1/2 & 0 \\ 0 & 4 \end{bmatrix}$ and $P^{-1} = \begin{bmatrix} 0 & 2 \\ 1 & -1.8 \end{bmatrix}$

Let

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -2 \end{bmatrix}, \quad B = \begin{bmatrix} 4 & -6 \\ 0 & -1 \end{bmatrix}, \quad C = \begin{bmatrix} 0 & 7 \\ -2 & 0 \end{bmatrix}$$

and

$$\tilde{C} = P^{-1}A^{-1}C = \begin{bmatrix} 2 & 0 \\ -1.8 & 7 \end{bmatrix}$$

Is there a locally unique solution? Solve for $\hat{C}_0$ and $\hat{I}_0$ when $\hat{G}_0 > 0$ with $\hat{G}_t = 0$ for $t \geq 1$

- $n_u = n_y = 1$
- $R_{uy} = -1.8$ is invertible
- $|\lambda_i| \neq 1$

So yes there is a locally bounded solution.